

ASML Holding N.V. (NASDAQ:ASML) — Equity Analysis

Analysis date: 2026-10-04 · **Mode:** Standard · **Sector playbook:** semiconductors (equipment sub-sector)
Basis: Consolidated · US GAAP · € millions unless stated (ASML reports in euros; the shares Santi buys trade in US dollars on Nasdaq) **Latest reported period:** Q2 2026 (quarter ended 2026-06-28), published 2026-07-15 [6-K, 2026-07-15] **Price reference:** \$1,867.31 (Nasdaq close 2026-10-02, data/prices/ASML.csv) = about €1,658.8 at EUR/USD 1.1257 (Yahoo Finance EURUSD=X close, 2026-10-02) · **Market cap:** about €638.5bn [E: 384.9m diluted shares, Q2 2026 6-K × €1,658.8] · **EV:** about €633.9bn [E, valuation.py]

Plain-English summary (for Santi)

The one-liner: ASML is the only company in the world that makes the machines needed to print the most advanced computer chips. Business is booming because of AI. But the share price already assumes many more years of fast growth, so the risk is less "is this a great company?" (it is) and more "have I paid too much for it?".

What it sells. Chips are made by shining light through a stencil onto a silicon disc, a bit like a slide projector printing a pattern. The smaller the pattern, the more powerful the chip. ASML builds those "projectors" (lithography machines \[machines that print chip patterns with light\]). Its top model uses extreme ultraviolet light (EUV \[a very short wavelength of light that can print tiny details\]). Nobody else sells EUV machines [20-F FY2025, Business section]. Each one is the size of a bus.

Who buys and how it makes money. A handful of chipmakers (TSMC, Samsung, Intel, SK Hynix, Chinese fabs). Four customers each bought more than 10% of everything ASML sold in 2025, together 61.2% [20-F FY2025, geographic/segment note]. About 75% of sales are new machines and 25% is servicing and upgrading machines already installed [20-F FY2025, Item 5: €24,474.3m systems vs €8,193.0m service].

What the numbers say. 2025 sales were €32.7bn, up 15.6%, and it kept about 29 cents of profit from each euro of sales [20-F FY2025]. In July it raised its 2026 forecast to €43–45bn [6-K, 2026-07-15], from €36–40bn three months earlier [6-K, 2026-04-15]: AI chip demand is pulling orders forward.

Case for: a monopoly in the most important step of chipmaking, high profits, net cash, and a growing pile of service revenue. **Case against:** the price. At about 60 times the last 12 months of profit, versus 34–37 times at the end of each of the last four years, today's price assumes cash flow grows about 19% a year for ten years (our estimate). Also: 29% of 2025 sales went to China, which governments can restrict at any time.

Biggest open risk right now: a proposed US law (the MATCH Act) could stop ASML selling older machines to China and servicing those already there. It is not law, and its status is unconfirmed (tier-3 reports only).

What would prove the thesis wrong: the 2026 forecast cut, gross margin falling back below 50%, or new export rules that block sales to China.

New words in this deep dive: *P/E* (share price ÷ yearly profit per share: how many years of today's profit you pay for); *reverse DCF* (working backwards from the price to the growth it assumes); *gross margin* (what is left of each euro of sales after the cost of making the product); *free cash flow* (cash left after running and investing in the business); *contract liabilities* (customer deposits for machines not yet delivered); *export controls* (government rules on what may be sold to which countries).

REGENCY STATEMENT

Most recent reported period incorporated: Q2 2026, published 2026-07-15 (6-K). Events checked through: 2026-10-04 (filings list in data/filings/2026-10-04.json; holding-researcher file data/research/2026-10-04/ASML.json when available). Material events since the last full-year data: pending US MATCH Act (China export curbs, unconfirmed status); TSMC said 2026-04-22 it has no plans to adopt High-NA EUV through 2029 (Investing.com, tier 3); guidance raised twice (April and July 2026); auditor changed from KPMG to PwC for FY2025; capacity plan of about +30% EUV for 2027. Any invalidation trigger already tripped at the time of writing: no. Next scheduled event: Q3 2026 results on 2026-10-14, 07:00 CEST [asml.com investor calendar, per search result 2026-10-04].

DATA QUALITY NOTE

Item	Statement
Primary sources	Form 20-F FY2025, accession 0001628280-26-011378 (filed 2026-02-25); 6-K Q2 2026 press release and US GAAP financial statements, accession 0001628280-26-048235 (2026-07-15); 6-K Q1 2026 press release, accession 0001628280-26-025147 (2026-04-15); SEC XBRL company facts (CIK 937966).
Secondary sources	Yahoo Finance for the share price and EUR/USD rate only. Peer figures: SEC XBRL company facts for Applied Materials (10-K FY2025), Lam Research (10-K FY2026), KLA (10-K FY2026).
As-of dates	Financials to 2026-06-28. Price 2026-10-02 close.
Consolidated vs standalone	Consolidated throughout.
Currency and units	€ millions unless stated. USD converted at 1.1257 (2026-10-02).
What is estimated	[E] market cap and EV (diluted weighted shares, not period-end count); [E] Q2 2026 short-term debt (FY2025 current borrowings €1,681.9m minus €692.9m repaid in Q1 2026); [E] EBITDA uses FY2025 D&A €1,025.9m as a TTM proxy; [TTM] = Q3 2025 to Q2 2026; scenario EPS and multiples are analyst judgements.
What is missing	Order bookings (ASML stopped publishing them in 2026); backlog only as €38.8bn in the 20-F CEO letter (as-of date not stated); EUV vs DUV sales split for 2026; peer market values (no peer prices pulled); KLA gross and operating profit (XBRL tags not found).
Known comparability issues	ASML reports calendar years; Applied Materials' year ends October, Lam's and KLA's June. Peers are US GAAP in USD; ratios compared, not absolute sizes.

VERDICT AND KEY RISKS

Verdict: An exceptional, monopoly-protected business in a strong upswing, priced for many more years of fast growth. Business quality high; valuation stretched against its own history; margin of safety thin. Confidence: high on the business facts, low on any valuation call.

Composite score: 7.1 / 10 (score.py, semiconductors preset, unedited default benchmarks) · **Sector playbook applied:** semiconductors · **Situation flags:** cyclical upswing (equipment orders follow chipmaker spending)

The three things that matter most:

1. **Monopoly in EUV.** Sole supplier; 48 EUV systems recognised in 2025 vs 44 in 2024 [20-F FY2025, KPIs]. Gross margin 52.8% in 2025, 54.0% in Q2 2026 [20-F; 6-K 2026-07-15].
2. **Demand is accelerating.** 2026 sales guidance €43–45bn [6-K 2026-07-15] vs €32,667.3m in 2025 [20-F], about +32% to +38% [computed]. Capacity for 2027 planned about 30% higher for low-NA EUV and DUV immersion [6-K 2026-07-15].
3. **The price assumes a lot.** Trailing P/E 60.0x [valuation.py: €1,658.8 / TTM diluted EPS ~€27.5], against 34.3x–37.3x at the last four year-ends [computed from 20-F share price table and diluted EPS]. Reverse DCF: ~18.9% a year FCF growth for 10 years at a 9% discount rate [valuation.py].

Key risks — what could break this:

1. **Valuation compression.** If the P/E went back to its 2022–2025 year-end range (34–37x) with profits unchanged, the price would fall roughly 40% [computed: $35/60 - 1$]. No fundamental change is needed for this.
2. **China / export controls.** China was €9,519.7m, 29.1% of 2025 sales [20-F FY2025, geographic note], and about 20% of 2026 sales per management [Q2 2026 call, search summary]. Dutch and US licence rules decide what can ship. A pending US bill, the MATCH Act (introduced 2026-04-02), would reportedly push allies to stop sales of older DUV immersion machines to major Chinese chipmakers *and* servicing of machines already installed there; it was reported in August as heading into the Senate defence bill, and the Dutch prime minister said on 2026-09-25 he was "very confident" of a deal [NL Times, 2026-08-20; De Belegger; DIGITIMES 2026-09-25 — all tier 3, not confirmed by a primary source]. Status in October 2026: not law; unconfirmed.
3. **Customer concentration and cycle.** Largest customer 23.9% of 2025 sales, top two 38.0% [20-F FY2025, risk factors]. If one big chipmaker pauses spending, orders fall fast: equipment is the most cyclical part of the chip chain [semiconductors playbook].

What would change the verdict: see Section 10.

1. The Business — What It Sells and How It Makes Money

- **What customers buy and why ASML.** Lithography systems: EUV (NXE low-NA and EXE high-NA models) for the finest chip layers, and DUV (deep ultraviolet) immersion and dry systems for the rest. Plus metrology and inspection tools, and "installed base management": service contracts and upgrades. Chipmakers choose ASML because, at the leading edge, there is no alternative EUV supplier [20-F FY2025, Business].
- **Revenue build.** Units × price, plus service. 2025: 327 lithography systems sold (418 in 2024), of which 48 EUV [20-F FY2025, KPIs]; 4 EXE (high-NA) and 44 NXE systems recognised [20-F FY2025, Item 5]. Fewer units but higher sales means the mix shifted to expensive EUV machines.

- **Mix.** Net system sales €24,474.3m (74.9%), +12.4%; service and field options €8,193.0m (25.1%), +26.2% [20-F FY2025, Item 5 results table]. Service grows as the installed base grows, and it is steadier than new-machine sales.
- **Where the money goes.** Cost of sales 47.2% of sales, R&D 14.4%, SG&A 3.9% → operating margin 34.6% [20-F FY2025, Item 5]. R&D of €4,698.8m is mostly fixed, so higher sales lift margins (operating leverage).
- **Cash path.** Customers pay large deposits years ahead: contract liabilities (money received for machines not yet delivered) were €19,372.6m at end-2025 [20-F XBRL, ContractWithCustomerLiability]. This makes cash flow lumpy: €11,410.3m operating cash in Q4 2025, then –€2,185.6m in Q1 2026 [6-K 2026-07-15, quarterly summary].

2. Sector Classification and Playbook Applied

Sector / sub-sector: Semiconductor equipment (wafer-fab equipment, lithography). **Playbook:** references/sectors/semiconductors.md.

Why: ASML sells tools to fabs; its earnings follow chipmakers' capital spending with a lag, which is the equipment pattern in the playbook.

Standard metric	Status here	Why
EV/EBITDA	Use with care	ASML is asset-light (capex 4.8% of 2025 sales); D&A is small, so EV/EBITDA ≈ EV/EBIT. Shown, but P/E and FCF yield lead.
ROIC / ROCE	Distorted	Customer prepayments (€19.4bn) shrink invested capital, making ROIC look extreme. Reported as an estimate only.
Current ratio	Not meaningful	Contract liabilities inflate current liabilities; they are prepaid orders, not stress.
Inventory days	Use with care	271 days [computed: €11,429.3m / €15,409.3m × 365, FY2025]; machines take months to build, so high inventory is structural.

Sector metrics used instead: EUV units, system vs service mix, gross margin trend, contract liabilities, customer and country concentration.

3. Situation Classification

Situation: Cyclical company in an upswing. Equipment orders are "the derivative of the derivative" of chip demand [semiconductors playbook]. **Implication:** today's earnings are not mid-cycle; a P/E on peak-ish earnings understates risk. Scenarios below include a downturn case.

4. Scorecard

Category	Score /10	Weight	Weighted	One-line rationale
Business quality & moat	7.5	16%	1.19	EUV monopoly (9/10) offset by 61.2% four-customer concentration
Profitability	9.4	16%	1.49	34.6% operating margin; returns very high [E]
Growth	8.5	16%	1.36	R&D 14.4% of sales, sustained; capex 4.8%
Earnings quality	5.7	12%	0.68	Cash/profit 1.17x over 2023–25 (strong) but inventory days 271 (structural)
Balance sheet	10.0	10%	1.00	Net cash at end-2025 and Q2 2026
Management & governance	9.2	8%	0.73	Share count down 2.3% since 2022 via buybacks
Valuation	1.9	14%	0.26	P/E 60x, EV/EBITDA 47x, FCF yield 1.5%
Risk	5.0	8%	0.40	China/Taiwan exposure high; Zeiss single-source optics
Composite		100%	7.1	

Weighting rationale: score.py default semiconductors preset, unchanged. **Scoring basis:** shipped benchmark bands (indicative defaults, not edited for this run); 5 = sector average. Coverage: 92% of categories, 30% of metric weight; the balance sheet and governance categories rest on one metric each (fragile). **Gates performed:** red-flag gates in score.py raised none; auditor opinion and internal-control opinion were checked manually (Section 8).

5. Core Analysis by Dimension

5.1 Business Quality and Moat

EUV: sole supplier [20-F FY2025]. The moat is technology plus a supply chain no rival has: optics come exclusively from Carl Zeiss SMT, in which ASML holds a 24.9% stake [20-F FY2025, equity method note and risk factors]. Gross margin rose from 51.3% (2024) to 52.8% (2025) [20-F] and 54.0% in Q2 2026 [6-K] while volumes of cheaper systems fell, consistent with pricing power at the top end (inference, medium confidence).

5.2 Earnings Quality

Metric	Value	Own history	Peer median	Read
CFO / net income, cumulative 2023–25	1.17x [computed: 2023 alone 0.69x; 29,268.1 / 25,020.0, 2024 1.47x; 2025 1.32x 20-F XBRL]		n/d	Cash exceeds profit over 3 years
Effective tax rate	17.3% H1 2026 [6-K]	n/d	n/a	Below statutory (innovation inference) Dutch 25.8% box,
Share-based pay / sales	0.6% [computed: 202.3 / 32,667.3, 0.3% (2022) FY2025]		n/d	Small
Receivables	€7,252.6m at Q2 2026 vs €3,023.0m €4,996.9m Q2 2025 at end-2025 [6-K]	n/a		Rose while H1 cash flow was negative (–€482.5m); watch in Q3

Read: quality is good over a full year; quarters are noisy because deposits arrive in lumps. The H1 2026 swing (cash out while profits rose) is the one item to check in the Q3 report.

5.3 Balance Sheet

End-2025: cash €12,916.0m, debt €4,390.9m (€2,709.0m long-term + €1,681.9m current) [20-F]. Q2 2026: cash + short-term investments €7,581.5m, long-term debt €1,984.4m [6-K]; net cash about €4.6bn [E]. Equity €21,825.4m [6-K]. No leverage concern.

5.4 Cash Flow

FCF (CFO – PP&E – intangibles): 2022 €7,167.5m; 2023 €3,247.2m; 2024 €9,083.1m; 2025 €11,027.3m [computed from 20-F XBRL; 2025 matches the €11.0bn ASML reports]. TTM to Q2 2026: about €9,892.7m [TTM, computed]. Capex €1,573.6m in 2025 vs D&A €1,025.9m [20-F]: still investing above depreciation (the playbook's healthy sign). Returned €8.5bn to shareholders in 2025 (€5.95bn buybacks) [20-F].

5.5 Returns on Capital

ROE: 2023 70.4%, 2024 47.4%, 2025 50.5% [computed: net income / average equity, 20-F XBRL]. ROIC is not cleanly measurable because customer prepayments fund much of the working capital; the scorecard uses a conservative 45% [E]. Either way, returns are far above any reasonable cost of capital.

5.6 Growth

Sales: €13,978.5m (2020) → €32,667.3m (2025), about 18.5% a year [computed, 20-F XBRL]. Diluted EPS: €14.13 (2022) → €24.71 (2025) [20-F XBRL], helped by buybacks (diluted shares 398.0m → 388.9m). 2026 guidance €43–45bn [6-K 2026-07-15]. All organic.

6. Peer Comparison

Peer set: Applied Materials (AMAT), Lam Research (LRCX), KLA (KLAC): the largest US wafer-fab-equipment makers. **Why:** same customers and cycle. **Excluded:** Tokyo Electron (Japanese GAAP, no SEC XBRL), Canon/Nikon (lithography is a small part of their sales). **Caveat:** none of the peers

sells EUV, so ASML's position is structurally different; fiscal years differ (see Data Quality Note).

Metric	ASML (FY2025)	AMAT (FY to Oct-2025)	LRCX (FY to Jun-2026)	KLAC (FY to Jun-2026)	Peer median
Revenue (local m)	€32,667	\$28,368	\$23,233	\$13,579	—
Gross margin	52.8%	48.7% [13,808/28,368]	50.5% [11,725/23,233]	n/d	49.6%
Operating margin	34.6%	29.2% [8,289/28,368]	35.3% [8,200/23,233]	n/d	32.3%
R&D / sales	14.4%	12.6% [3,570/28,368]	10.2% [2,376/23,233]	11.3% [1,532/13,579]	11.3%
Capex / sales	4.8%	8.0% [2,260/28,368]	4.2% [966/23,233]	2.8% [376/13,579]	4.2%
P/E (trailing)	60.0x	n/d	n/d	n/d	n/d

Sources: each company's latest 10-K via SEC XBRL company facts (AMAT accn 0001628280-25-056742; LRCX 0000707549-26-000037; KLAC 0000319201-26-000027). **Read:** ASML's margins sit at or above the top of the group and it spends more on R&D, consistent with the monopoly. Peer valuations were not gathered, so "expensive vs peers" is not established here; "expensive vs its own history" is.

7. Valuation

Method: reverse DCF cross-checked against ASML's own P/E history. **Why:** stable, highly cash-generative, asset-light; FCF is meaningful. **Inputs:** base FCF €11,027.3m (FY2025, less lumpy than TTM); discount rate 9% (analyst assumption: no euro risk-free rate was sourced this session, so treat it as a round, conservative-ish figure); terminal growth 3%; 10-year stage 1.

Reverse-DCF: what the current price implies

At €1,658.8 (about \$1,867), the market is pricing in roughly **18.9% a year free-cash-flow growth for ten years**, then 3% forever [valuation.py]. **Reality check:** FCF grew about 15.4% a year 2022–2025 (€7,167.5m → €11,027.3m, computed) and sales about 18.5% a year 2020–2025. So the price asks for slightly *more* than ASML's very good recent record, for twice as long, with no downturn. Possible given AI demand; not a margin of safety.

Scenario table (estimates, not forecasts)

Scenario	Probability [E]	Key assumptions (2027)	Value / share	vs €1,658.8
Bear	25%	Chip-spending pause: sales ~€40bn, net margin 28%, EPS ~€29.3, P/E 25x	€733	–56%
Base	50%	Sales ~€52bn, net margin 32%, EPS ~€43.7, P/E 35x (own year-end average)	€1,530	–8%

Scenario	Probability [E]	Key assumptions (2027)	Value / share	vs €1,658.8
Bull	25%	Sales ~€60bn, net margin 34%, EPS ~€54, P/E 40x	€2,160	+30%
Probability-weighted	100%		€1,488	-10%

Probabilities and assumptions are judgements; the base case keeps today's growth story intact and only lets the P/E return to its own average. Ranges are wide on purpose: this is not a price target.

8. Red Flags and Governance

#	Flag	Severity	Evidence	Why it matters
1	Customer & country concentration	Med	Four customers 61.2%; China 29.1%, Taiwan 25.5% of 2025 sales [20-F FY2025]	Policy or one customer's capex decision moves revenue
2	Single-source optics (Zeiss)	Med	Exclusive arrangement; ASML owns 24.9% [20-F FY2025]	A Zeiss disruption would halt shipments
3	Lumpy cash / receivables jump	Low	H1 2026 CFO -€482.5m; receivables €7,252.6m vs €3,023.0m [6-K]	Timing of deposits; check it reverses
4	Auditor change	Low	PwC replaced KPMG (auditor 2015–2024) for FY2025 [20-F, auditor report and fees appendix]	Routine rotation is likely (Dutch rules), but noted

Checked and clear: auditor opinion unqualified for FY2025 (PwC) and internal control over financial reporting judged effective as of 2025-12-31 [20-F, Report of Independent Registered Public Accounting Firm]; no restatement found; no dilution (share count falling); SBC small; net cash.

9. The Bear Case

The bull case needs three things to stay true at once: AI chip demand keeps rising for years, China keeps buying at about a fifth to a third of sales, and investors keep paying about 60 times earnings. History says at least one usually breaks. Chip equipment is the most cyclical link in the chain: when chipmakers have built enough capacity they cut orders sharply, and ASML's 2024 (sales +2.6%, 20-F) shows how quickly growth can stall even for a monopoly. ASML is now adding about 30% EUV capacity for 2027 [6-K 2026-07-15], which is exactly what equipment makers do near cycle tops. China sales were inflated in 2024–2025 by stock-up buying ahead of restrictions; ASML itself expected China to fall back [Caixin Global, 2026-01-29, search summary only]. If growth merely slows to the 2022–2025 pace and the P/E returns to its own 35x average, the shares fall about 40% with no "bad news" at all. And with four customers at 61% of sales, ASML's growth is really TSMC's and Samsung's capex plan, which ASML does

not control.

Strongest counter to the bear case: the company raised 2026 guidance twice in four months and said on its Q2 2026 call that it is close to having all the 2027 orders it needs [Q2 2026 earnings call, 2026-07-15; transcript not read directly, so treat as medium confidence], so near-term demand is visible; and service revenue (25% of sales, +26%) cushions downturns.

10. Thesis-Invalidation Triggers

#	Trigger	Where to observe	By when	Action if hit
1	FY2026 sales cut below €43bn	Q3 6-K (2026-10-14) / Q4 6-K	Jan 2027	Thesis review: demand story weakening
2	Quarterly gross margin below 50%	6-K quarterly	Any quarter	Pricing power or mix weakening
3	MATCH Act (or a Dutch rule) enacted that stops DUV / immersion sales or servicing in China	US Congress record / ASML 6-K / Dutch government	NDAA late 2026 timetable,	Re-size China (about 20% of 2026 sales) in scenarios
4	Any single customer above 30% of annual sales	20-F FY2026	Mar 2027	Concentration risk rising
5	Operating cash flow negative for a full year	20-F FY2026	Mar 2027	Deposits drying up: orders slowing

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